

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id almr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

The screenshot shows the 'Email Header Analyzer' tool from MX Toolbox. It features a 'Paste Header:' section with a text area containing a sample email header. Below the text area is an 'Analyze Header' button. The header text includes delivery information, timestamps, and various email identifiers. At the bottom of the page, there is an 'ABOUT EMAIL HEADERS' section explaining the tool's purpose and a footer with copyright information.

Email Header Analyzer

Paste Header:

```
Delivered-To: tanakantojthn@gmail.com
Received: by 2002ac0c40f0b0:3fe:f9df:d043 with SMTP id t15csp615795imj; Thu, 20 Aug 2026 18:49:10 -0700 (PDT)
X-Received: by 2002a179021c43b0:2d5:2149:b314 with SMTP id d9443c01a7336-2d64ad9562cmr53569005ad.2.1787276950520;
Thu, 20 Aug 2026 18:49:10 -0700 (PDT)
ARC-Seal: i=f; a=rsa-sha256; t=1787276950; cv=none; d=google.com; s=arc-20260327;
b=gR2D+Rs03ofg4Epl3C0vEeUsl357+KcLWwCvPae1AWzGPh/aqbln79DRUSMQtbY
pKBu/cQWwGGZz+Y0X7EnGIBNc05a2YEzzPSs94m9nLWYvDU7W3bRwJthYhtG(SSB)
5lqqspekzG3J/HaakBnT215eEnQZXrJlMSiKl3wCwQouFXa5KZc7uNwCQu+LYeXo5d
giw/gxWHQkDN+6wvXqSxOJqVHe2TtFEs0Aev/BX9x18JTY7H7SV2GDbd2Czd57J8YF6u
BWobM/G3TPErYtL8EO4rbvbcKw9E+y+EKbLYMvzHGx2Yod+MW2Qbkw4GkTPJ+lhgEjw==
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, just read this tutorial.

Your IP is: 103.230.232.130 | Contact Terms & Conditions Security API Privacy Policy (590)-593-6002 | © Copyright 2004-2025 MXToolbox, Inc. All rights reserved. 149 Patents 10839363 62 & 11461736 B2

SPF and DKIM Information

dmARC:smaller1.binance.com

Show

Solve Email Delivery Problems

v=DMARC1;p=quarantine;rua=mailto:binance@rua.netcraft.com,mailto:dmarc@binance.com;ruf=mailto:binance@ruf.r

spf:mf.smaller1.binance.com:206.55.152.40

Show

Solve Email Delivery Problems

v=spf1 include:amazonses.com -all

dkim:smaller1.binance.com:qg7roxdubnbg5luukim7osq7u177gbyo

Show

Dkim Public Record:

p=MIIBIjANBgkqhkiG9w0BAQEFAADCAQ8AMIIBCgKCAQEaunMA19n6fyMcdAtLgr0ip1EE/j3T7FNKgnCAyC33IoIV6FO0I3hEpI1sreJ3;

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=qg7roxdubnbg5luukim7osq7u177gbyo; d=smaller1.binance.com;

dkim:amazonses.com:zh4gjftm6etwoq6afzugpky45synznly

Show

Dkim Public Record:

p=MIGfWA0GCSqG5Ib3DQEBAQUAA4GNADCBiQKBgQC1VvH5KXmXOHILciVUFkEz3TR/a1/p+tGdKXdw+e6ySidBHnsaLaFeR9iwj60SnmjVf

Dkim Signature:

v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=zh4gjftm6etwoq6afzugpky45synznly; d=amazonses.com; t=178;

Headers Found

IP is 100.238.220.130 | Contact Terms & Conditions Security AD Privacy Policy (888) 496-6052 | © Copyright 2004-2026, MXToolbox, Inc. All rights reserved. LG Patents 10039353 B2 & 11481796 B2

MX TOOLBOX[®]

SUPERTOOL

Login

Header Analyzed

Email Subject: Important: Transactions with Certain Crypto Platforms

Analyze New Header

Delivery Information

DMARC Compliant

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received Delay:

0 seconds

From h152-40.smtp-out.ap-northeast-1.amazonses.com to mx.google.com

to

0 0.2 0.4 0.6 0.8 1 1.2

Relay (Seconds)

Ho p	Delay	From	By	With	Time (UTC)	Blacklis t
1	*	h152-40.smtp-out.ap-northeast-1.amazonses.com 206.55.152.40	mx.google.com	ESMTP S	8/21/2026 1:49:10 AM	✓
2	0 second s		2002:ac0:c40f:0:b0:3fe:19df:d043	SMTP	8/21/2026 1:49:10 AM	

Headers Found

Header Name	Header Value
Delivered-To	tanakantnithin@gmail.com
X-Received	by 2002-a7-902fc43b0-2d5-2f49-b3f4 with SMTP id d9443c01a7336-2d64ad9562cmr53569005ad.2.1787276950520; Thu, 20 Aug 2026 18:49:10 -0700 (PDT)
ARC-Seal	i=1, a=rsa-sha256, t=1787276950, cv=none, d=google.com, s=arc-20260327, b=q82D+R303ofg4Epl3C0vEeUsI357+KcJyWcVtPgetAWzGPh/aqbtjq79DRUSM0tBY nKRu/cOWwGGZ+Y0X7EnGIRNcO6a2YEzzPSs94m9nL/YfyDU7W3bRwLthHYhtjGf55R 5kqqspekzG3J/HAakqBpI2f5eEpQ7Xr1JlMSKl3wCxOguFXa5KZ07uNmcOw+FYuXg5d giw/gxWlOKDN+6wXqSxOJqVH e2TTFs0Aev/RX9x18JTY7H7SV2GDbd2CrgR7J8YE6u BWnB/M/G3TPFRytl.8E04rbsvbcKw9F+y+FKbLYMvzHGx2Ynd+MWI2Ohkv4GkTPj+Hhg Fvaw==
ARC-Message-Signature	i=1, a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, h=feedback-id:date:message-id:list-unsubscribe-post: list-unsubscribe:content-transfer-encoding:mime-version:subject:to:from :dkim-signature:dkim-signature, bh=x9oULU/YhgfhCWLHQ4aKDS1mLH663CTK0hDTNJO4AA=, fh=gmlfMSmnGplUDajadY1vQmh68eNOZrOQwRbhdO9wHg=, b=BAUj1ptvw/6aD7Hf6szZYhRHngWP9HHfdAehFDcaVgwRyNAHy0/gXds6tmQVoz6pF izK5MXHI+LDhVAo3o67Q+ake2FC6Z7FWj5KSo6FJ Q3dLhaZC23XhN48jb9M8l6fwzph +F2mJVwyh2l8WVyl+Ar5Ztb4FLUr8nQegnR8F7XJNV TernaeKlorp/EHHdrkVJE1VMi Kpo wcNuUcMTHagxL1oqYrN8uwtfmkwpFMU0Q3SvOjqAKy/x2lRy7B6wLXVYac2w0y ed+U50woJ2zxqD2L2ayIKbF/8LAIMt1Q cmGT/k5aXoAnXlVnSJBg+cs7VR05aXW8aUlin E4Jg==, dara=google.com
ARC-Authentication-Results	i=1, mx.google.com; dkim=pass header.i=@smaller1binance.com header.s=qq7roxdubnbg5luukim7osq7ui77gbyo header.b=c5zs1ss; dkim=pass header.i=@amazonises.com header.s=zh4qjtm6etwo6afzupgky45synzly header.b=SGTJKsFz; spf=pass (google.com: domain of 010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com designates 206.55.152.40 as permitted sender) smtp.mailfrom=010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) headerfrom=smaller1binance.com
Return-Path	<010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com>
Received-SPF	pass (google.com: domain of 010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com designates 206.55.152.40 as permitted sender) client-ip=206.55.152.40;
Authentication-Results	mx.google.com; dkim=pass header.i=@smaller1binance.com header.s=qq7roxdubnbg5luukim7osq7ui77gbyo header.b=c5zs1ss; dkim=pass header.i=@amazonises.com header.s=zh4qjtm6etwo6afzupgky45synzly header.b=SGTJKsFz; spf=pass (google.com: domain of 010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com designates 206.55.152.40 as permitted sender) smtp.mailfrom=010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@mfsmailler1binance.com; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) headerfrom=smaller1binance.com
DKIM-Signature	v=1, a=rsa-sha256, q=dns/txt, c=relaxed/simple, s=qq7roxdubnbg5luukim7osq7ui77gbyo, d=smaller1binance.com, t=1787276949, h=From:To:Subject:MIME-Version:Content-Type:Content-Transfer-Encoding:List-Unsubscribe:List-Unsubscribe-Post:Message-ID:Date, bh=PTGJqfAht6z7mhVR2ty9l2W3FQg2K7z6Gguo7A=, b=c5zs1ss/v/98EkWkx0Tw2f4QSNBhwDhCYkOXUE9xKhX4hviMnVLtwrUMp6erK5 rLMCON50ToZixFrUjD/buOmoNHmZNYlqHoWHKHQP0KRknQmt5KUSDSNHSa/PDTJW280 yH AEYMUU2PzBk817MU8efZhpvOBj6lg2+ cZlYDzmzR4tqTxElRV046x+ Qk/3cOAYt LMWnTuLISKfz2bg2YSHVwmDORD9btOn3SA7w2mfnD/gFPwoS9ESL94s9bJiHnM3oHC JDLEotYlvKQtcYuOqr7nJpr1X81OmzaGF72hQH/lp9TsSKYj/fJuKuQEWaxo8dLZe K1ApM86aHg==
From	Binance <do_not_reply@smaller1binance.com>
To	tanakantnithin@gmail.com
Subject	Important: Transactions with Certain Crypto Platforms
MIME-Version	1.0
Content-Type	text/html; charset=UTF-8
Content-Transfer-Encoding	quoted-printable
List-Unsubscribe	<https://www.binance.com/bapi/growth/v1/public/orion/marketing-message/user-subscription/unsubscribe?param=EywG3vu0E5kL4rdfn9J%2B0TgnfYlOmF2RCD%2B%2B8bn%2FgrTNrOLTUwEnob4ENOOCc9WoiKwZ8jt8HByWp0iUlgqeq5v8/%2BY%2BLdwVPfwRtc6t4%2BXN45j/%2FcB9Jufkg8X4uLgyrSbb/%2B3UIAI2R7p/%2FqFegKIYwv7AsXP9xhZaJhJLkZxzE4oyJWYtCoXi894Z3gORTs/%2FQ%2BCTZLbX7R9SfdJxRKRQMe%2FAZm56XmPWbjRTbrOu7qF7j7sOMkO3EoKp%2BEJAp55HiuQnIY%2BgDtrt8hJMXz8p5h/mPqJMJQMkdfsnNbG6oMZKbSM13y9Tb%2Blg05mBYUApv%2FJ5lox5TsPHPd0QGFHFwJpEMvhOXT18XRkXqj7RB2EDc>
List-Unsubscribe-Post	List-Unsubscribe=One-Click
Message-ID	<010601a0220f8bc-bf1f58c0-b66b-439d-8a2c-62d19729407f-000000@ap-northeast-1.amazonaws.com>
Date	Fri, 21 Aug 2026 01:49:09 +0000
Feedback-ID	:f1ap-northeast-1+N/3U6Xakq/FV4PhYfc6R5xuBpQbOdoFgPgHzDfXZ7s=AmazonSES
X-SES-Outgoing	2026.08.21-206.55.152.40

Received Header

Delivered-To: tanakantinithin@gmail.com
Received: by 2002:ac0:c40f:0:b0:3fe:19df:d043 with SMTP id t15csp615795lmj; Thu, 20 Aug 2026 18:49:10 -0700 (PDT)
X-Received: by 2002:a17:902:fc43:b0:2d5:2f49:b3f4 with SMTP id d9443c01a7336-2d64ad9562cmr53569005ad.2.178727695052
ARC-Seal: i=1; a=rsa-sha256; t=1787276950; cv=none; d=google.com; s=arc-20260327; b=q82D+Rs03ofg4Eplx3C0vEeUsI357+K
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=feedback-id:date:mesa
ARC-Authentication-Results: i=1; mx.google.com; dkim=pass header.i=@smailler1.binance.com header.s=qg7roxdubnbg5luuk
Return-Path: <010601a02201f8bc-1b1158c0-b66b-439d-8a2c-62d19729407f-000000@mf.smailler1.binance.com>
Received: from h152-40.smtp-out.ap-northeast-1.amazonaws.com (h152-40.smtp-out.ap-northeast-1.amazonaws.com, [206.5
Received-SPF: pass (google.com: domain of 010601a02201f8bc-1b1158c0-b66b-439d-8a2c-62d19729407f-000000@mf.smailler1.
Authentication-Results: mx.google.com; dkim=pass header.i=@smailler1.binance.com header.s=qg7roxdubnbg5luukim7osq7ui
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=qg7roxdubnbg5luukim7osq7ui77gbyo; d=smailler1.bina
DKIM-Signature: v=1; a=rsa-sha256; q=dns/txt; c=relaxed/simple; s=zh4gjftm6etwoq6afzupky45synznl; d=amazonses.com
From: Binance <do_not_reply@smailler1.binance.com>
To: tanakantinithin@gmail.com
Subject: Important: Transactions with Certain Crypto Platforms
MIME-Version: 1.0
Content-Type: text/html; charset=UTF-8
Content-Transfer-Encoding: quoted-printable
List-Unsubscribe: <https://www.binance.com/bapi/growth/v1/public/orion/marketing-message/user-subscription/unsubscr
List-Unsubscribe-Post: List-Unsubscribe=One-Click
Message-ID: <010601a02201f8bc-1b1158c0-b66b-439d-8a2c-62d19729407f-000000@ap-northeast-1.amazonaws.com>
Date: Fri, 21 Aug 2026 01:49:09 +0000
Feedback-ID: ::1:ap-northeast-1.+N/3U6Xakq/FV4PnYfc6R5xuBp6b0doFgPgHzdFXZ7s=:AmazonSES
X-SES-Outgoing: 2026.08.21-206.55.152.40

<html xmlns=3D"http://www.w3.org/1999/xhtml" xmlns:v=3D"urn:schemas-microso=
ft-com:vml" xmlns:o=3D"urn:schemas-microsoft-com:office:office"><head> <tit=
le> </title> <!--[if lms]><!--> <meta http-equiv=3D"X-UA-Compatible" conte=
nt=3D"IE=3Dedge"> <!--<![endif]>> <meta http-equiv=3D"Content-Type" conten=
t=3D"text/html; charset=3DUTF-8"> <meta name=3D"viewport" content=3D"width=
=3Ddevice-width, initial-scale=3D1"> <style type=3D"text/css"> #outloo=
k a { padding:0; } body { margin:0;padding:0;-webkit-text-size-adjust:=

